

Beyond the Average Score: Within-Model and Between-Model Judgment Variability in LLM Panels for Startup Idea Evaluation

First Author

Department of Computer Science and Engineering, [Your Institution]

[City, Country]

author@your-institution.edu

Abstract— Automated startup idea evaluation increasingly relies on large language models (LLMs) as judges, typically reporting a single aggregate score per idea. This practice assumes that disagreement among evaluators is noise to be averaged away. Prior work shows that inter-judge disagreement in LLM evaluation panels is frequently structured rather than random, but this finding has largely been established in domains with comparatively assessable ground truth, such as natural language inference, exam grading, and video-quality judging. Startup idea evaluation differs in at least three respects: no ground truth exists even in principle at evaluation time, pitch text is deliberately persuasive rather than neutral, and real, well-known companies risk being recognized from model pretraining rather than reasoned about fresh. This paper presents a controlled panel study that separates within-model judgment variability (repeated evaluations of the same idea by one model under fixed decoding parameters) from between-model disagreement (a panel of independent models evaluating the same idea once each), using a fixed rubric, a strict structured output schema, and a set of stability, agreement, and panel-spread metrics computed on raw judgments rather than on the aggregate score alone. [N] startup ideas drawn from [dataset source] were evaluated using [k] LLMs as panel judges, run at temperature 0.0 with $R = 10$ same-model replicates. Results are reported in terms of flip rate, self-consistency rate, score standard deviation, mean pairwise absolute deviation (MPAD), verdict entropy, and effective viewpoints, comparing cross-model disagreement against the same-model noise floor. [Result and impact statement to be completed following experiment execution.]

Keywords— *LLM-as-judge, panel disagreement, evaluation reliability, startup idea evaluation, inter-model agreement, judgment variability*

I. INTRODUCTION

Large language models (LLMs) are increasingly deployed as automated judges — scoring essays, grading code, ranking model outputs, and, more recently, evaluating early-stage business ideas before any market traction exists. This last use case is attractive because manual expert review of startup pitches is slow, expensive, and difficult to scale, while an LLM can process a written idea description in seconds and return a structured verdict. As AI-assisted startup screening tools move from research prototypes toward practical decision support, it becomes important to understand not just whether an LLM's average judgment looks reasonable, but whether that judgment is reliable and what a single reported score is hiding.

The standard practice in this space, mirroring practice in LLM-as-judge evaluation generally, is to collapse a panel of evaluators — human or model — into a single aggregate score and to validate an AI judge by checking whether its average score correlates with the average expert score. This approach implicitly treats disagreement between evaluators as measurement noise. A growing body of work on LLM-judge panels shows this assumption does not generally hold: disagreement between judges is often structured — concentrated on specific, genuinely ambiguous items rather than spread uniformly — and averaging over it can obscure exactly the cases where an evaluator is least reliable. However, this line of work has mainly been demonstrated in domains where a comparatively stable notion of correctness exists at evaluation time (e.g., inference labels, exam rubrics, video quality). Startup idea evaluation lacks that property even in principle: whether an idea succeeds depends on future market conditions, competition, execution, and luck that are unknowable when the idea is judged.

This paper proposes and executes a controlled panel study that distinguishes within-model judgment variability from between-model disagreement for the specific task of pre-launch startup idea evaluation. Rather than treating either as automatically meaningful, both are computed under matched, fixed experimental conditions (fixed prompt, fixed rubric, fixed decoding temperature, fixed output schema) and compared directly. The study further examines whether observed disagreement is better explained by three mechanisms plausibly specific to this domain: the absence of any eventual ground truth, the persuasive framing typical of pitch text, and memorization/contamination risk when real, recognizable companies are used as cases.

The contributions of this paper are:

- A controlled experimental harness that separates same-model replicate variability (C1) from cross-model panel disagreement (C2) under matched decoding and prompting conditions, with all raw and parsed outputs logged and preserved.
- A judgment-profile-based analysis (score, verdict, and a reduced rubric dimension set) that avoids collapsing multidimensional evaluator disagreement into a single scalar before analysis.

- A four-pattern interpretation grid relating within-model stability to between-model divergence, used to classify cases rather than report a single reliability number.
- An empirical demonstration of whether cross-model disagreement exceeds same-model noise across N startup idea cases, with an accompanying effective-panel-size estimate. [To be completed after experiment execution.]
- A discussion of whether startup evaluation's domain-specific properties (no ground truth, persuadability, memorization risk) plausibly account for observed differences from disagreement patterns reported in other LLM-judge domains.

II. LITERATURE SURVEY

Related work is organized into four groups: disagreement as an informative signal, LLM-judge reliability and instability, panel and multi-agent evaluation frameworks, and LLM-based startup or business idea evaluation. Table I summarizes representative studies by method, application, and key finding; limitations of each line of work motivate the gap addressed in Section II-E.

A. Disagreement as Informative Signal

Work on annotator disagreement in NLP and computer vision argues that collapsing multiple human judgments into one ground-truth label can erase systematic, meaningful differences of opinion rather than noise, particularly in subjective tasks such as hate speech detection and value-laden argument classification [1]–[3]. Surveys of this area report that, given sufficiently large datasets, training directly on soft or distributional labels typically outperforms training on a single aggregated label [3]. This literature establishes the general principle motivating the present study — disagreement can be signal — but is concerned with human annotators, not automated LLM judges.

B. LLM-Judge Reliability, Instability, and Bias

A rapidly growing body of work examines the reliability of LLMs used as judges. Studies report that identical repeated evaluations of the same input can produce materially different scores and verdicts, that this instability persists even at temperature zero due to serving-level nondeterminism, and that judges can be self-consistent yet diverge sharply from one another, reflecting stable but distinct evaluative dispositions rather than random error [4]–[8]. Other work documents concrete bias channels — position bias in pairwise comparisons, verbosity bias, and sensitivity to prompt wording and grading-scale choice — and argues that reporting practice should separate consistency, agreement, and bias rather than collapsing them into a single score [9]–[12]. A related critique observes that when judges are highly correlated, ensembles of nominally independent judges can behave as if they were only a small number of effectively distinct judges, undermining naive vote-counting or averaging [13].

C. Panel and Multi-Agent Evaluation Frameworks

Several frameworks propose replacing a single LLM judge with a panel or committee, on the premise that diverse judges reduce single-model bias [14]. Related work on multi-agent deliberation shows that additional agents or discussion rounds do not automatically improve outcomes and can waste computation without a controller that gates evidence quality and schedules disagreement productively [15]. Committee-based grading frameworks emphasize transparency and reproducibility of the panel's individual member outputs, rather than reporting only an arbitrated or averaged final grade [16].

D. LLM-Based Startup and Business Idea Evaluation

A smaller literature applies LLMs directly to startup evaluation and prediction, including multi-agent frameworks for founder-idea fit assessment, multitask models for structured business outcome reasoning, and memory-augmented in-context prediction of startup success [17]–[20]. Work in this space has found that (a) fine-grained expert agreement on business ideas can be near zero even when coarse agreement (e.g., top-half vs. bottom-half ranking) is comparatively high, (b) LLM predictive skill on startup survival can be partly explained by memorization of a company's digital footprint from pretraining rather than genuine reasoning about the idea, and (c) some LLM judges behave as "narrative funders," more swayed by qualitative storytelling than others. None of this literature specifically decomposes LLM panel disagreement on startup ideas into within-model and between-model components under controlled conditions.

E. Identified Gap

The general finding that LLM-judge panel disagreement is structured rather than purely random is reasonably well established [4]–[13]. What remains open is whether that finding holds the same way in a domain that differs from prior test-beds (NLI, exam grading, video quality) on three concrete mechanisms: the complete absence of an eventual ground truth, the persuasive construction of the input text itself, and a live memorization/contamination confound for recognizable real-world cases. This paper addresses that gap directly, using startup idea evaluation as the test domain.

TABLE I SUMMARY OF REPRESENTATIVE RELATED WORK

Group	Focus	Key Limitation
Disagreement as signal	Soft/distributional labeling from human annotator disagreement	Human annotators, not LLM judges
LLM-judge reliability	Instability, self-consistency, position/verbosity bias	Not evaluated in no-ground-truth domain
Panel/multi-agent frameworks	Committee grading, diverse-judge panels, deliberation control	Rarely decomposed within- vs. between-model
Startup/business evaluation	Founder-idea fit, outcome prediction, memory-augmented reasoning	Does not isolate within- vs. between-model

III. METHODOLOGY

A. System Overview

The experimental harness takes a dataset of startup idea cases and evaluates every case under two conditions. In Condition 1 (C1, same-model replicates), a single model evaluates each idea independently R times under fixed decoding parameters, establishing within-model judgment variability. In Condition 2 (C2, cross-model panel), k distinct models each evaluate every idea once, establishing between-model disagreement. Every evaluator call is conditionally independent: each judge sees only the idea text and a fixed evaluation prompt, with no access to other evaluators' outputs, no retrieval, and no memory across calls. Fig. 1 summarizes the end-to-end pipeline from dataset to reported metrics.

[Pipeline diagram: Case Dataset $\rightarrow$ Task Generator $\rightarrow$ Inference (C1 / C2) $\rightarrow$ Raw Run Logger $\rightarrow$ JSON Validator $\rightarrow$ Results Table $\rightarrow$ Metrics $\rightarrow$ Tables and Figures]

Fig. 1. End-to-end experimental pipeline from case dataset to reported metrics.

B. Data Collection

Dataset source: [YC public batch list / Kaggle startup-success dataset / researcher-authored synthetic cases]

- Number of cases: $N = []$
- Composition: $[]$ real cases (source: $[]$) / $[]$ synthetic cases
- Cases with known real-world outcome label available: $[]$ of N
- Domains represented: $[]$

C. Fixed Experimental Parameters

The following parameters were fixed across all runs and recorded in a machine-readable experiment manifest, since uncontrolled variation in any of them can independently change conclusions about disagreement (Table II).

TABLE II FIXED EXPERIMENTAL PARAMETERS

Parameter	Value
Primary temperature	0.0 (fixed across C1 and C2)
Robustness temperature	0.7, subset of highest-disagreement cases only
Prompt template	Single fixed template, identical across models/conditions
Output schema	Strict JSON: score (1–10), verdict, confidence (0–100), reasoning (≤ 60 words), rubric dims. (Sec. III-D)
Inference mode	Single-pass; no retrieval, tools, memory, self-correction
Caching	Disabled for all replicate (C1) calls
Model provenance	Exact name, snapshot/version, provider, endpoint per call

D. Evaluator Output Schema and Judgment Profile

Rather than analyzing only the final aggregate score, each evaluator call returns a judgment profile consisting of an overall score, a binary verdict, a confidence value, a short free-text rationale, and scores on a reduced set of three rubric

dimensions (market potential, technical feasibility, business viability), each on a 0–5 integer scale. The reduced dimension set was chosen to make criterion-level drift visible without the parsing and validation cost of a full multi-dimension rubric.

E. Metrics

All metrics are computed on schema-valid outputs only; the invalid-output rate is reported separately as its own reliability signal.

1) *Within-Model Stability (C1, R Replicates Per Case)*: the flip rate, self-consistency rate, score variance, and rubric-profile distance are computed as follows.

$$\text{FlipRate} = 1 - \max_c(n_c) / R \quad (1)$$

$$\text{SCR} = 1 - \text{FlipRate} \quad (2)$$

$$s^2 = (1/(R-1)) \sum (x_r - \bar{x})^2 \quad (3)$$

$$D(A,B) = \sum |A_i - B_i| / (n_{\text{dims}} \times 5) \quad (4)$$

where n_c is the count of verdict category c across R replicates, $\bar{x}$ is the mean score across replicates, and $D(A,B)$ is the mean absolute per-dimension distance between two rubric profiles, normalized to $[0, 1]$.

2) *Between-Model Disagreement (C2, k Judges Per Case)*: verdict entropy, mean pairwise absolute deviation, range spread, and effective viewpoints are computed as follows.

$$H = -\sum p_c \log_2(p_c) \quad (5)$$

$$\text{MPAD} = (2 / (k(k-1))) \sum_{a < b} |s_a - s_b| \quad (6)$$

$$\text{Range} = \max_j(s_j) - \min_j(s_j) \quad (7)$$

$$N_{\text{eff}} = k / (1 + (k-1) \times \bar{p}) \quad (8)$$

where p_c is the panel's verdict-category proportion on a given case, s_a and s_b are member scores, k is the nominal panel size, and $\bar{p}$ is the mean pairwise score correlation (Pearson) across panel members, used as a similarity measure in the Kish-style effective-viewpoints estimator. A pairwise model agreement matrix and a pairwise score correlation matrix are also computed across all cases.

3) *Reasoning-Level Similarity (Exploratory Only)*: free-text rationale strings are embedded with a lightweight sentence-embedding model and compared via cosine similarity across runs (C1) and across judges (C2). This is reported as a rough, exploratory signal of whether reasoning content is stable, not as a validated measure of conceptual agreement, consistent with literature cautioning that model-generated explanations can reflect surface heuristics rather than substantive judgment.

F. Four-Pattern Interpretation Grid

Cases, and the dataset as a whole, are classified according to the combination of C1 and C2 dispersion, following the interpretation grid in Table III.

TABLE III INTERPRETATION GRID FOR WITHIN- VS. BETWEEN-MODEL DISPERSION

C1	C2	Interpretation
Low	Low	Stable task; judges converge

exist, whether disagreement aligns with outcomes that were themselves surprising or borderline. This analysis is exploratory given small N.]

F. Domain-Specific Mechanisms

[Discuss, with whatever evidence the data supports: (i) coarse- vs. fine-grained agreement split as evidence of the no-ground-truth mechanism; (ii) evidence of persuasion-susceptibility differences across models; (iii) evidence that real/recognizable cases behave differently from synthetic cases, consistent with a memorization mechanism. Where the data does not speak to a mechanism, this is stated explicitly rather than omitted.]

G. Comparison Against a Naive Aggregation Baseline

[Compare the disagreement-aware analysis against a naive baseline reporting only the mean panel score per case. State explicitly what information the naive baseline would have hidden, using the highest-entropy cases as concrete examples.]

V. CONCLUSION

This paper set out to test whether disagreement among LLM evaluators of pre-launch startup ideas behaves like disagreement already documented in other LLM-judge domains, or whether the specific properties of startup evaluation — the absence of an eventual ground truth, the persuasive framing of pitch text, and memorization risk for recognizable companies — produce a distinguishable pattern. Using a controlled harness that separates within-model judgment variability from between-model panel disagreement under matched decoding and prompting conditions, [key empirical finding to be stated here once the experiment has been run].

The practical implication is that automated startup evaluators should report panel dispersion alongside any aggregate score, and that high-dispersion cases should be routed for escalation rather than resolved by averaging.

Future work includes extending this harness with a human-expert evaluation layer for direct comparison against the LLM panel, a formal temporal-leakage audit to directly test the memorization mechanism rather than only naming it as a limitation, personalized per-evaluator modeling rather than a shared panel-level analysis, and mirrored-order or paraphrase-based bias audits of the kind used in other LLM-judge reliability studies.

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